

1. Database Commands

```
> use("admin")
< switched to db admin
> show dbs
< admin          48.00 KiB
  config         72.00 KiB
  local          72.00 KiB
  studentDB      72.00 KiB
> use studentDB
< switched to db studentDB
> db.dropDatabase()
< { ok: 1, dropped: 'studentDB' }
```

2. Collection Commands

```
> show collections
<
> db.createCollection("students")
< { ok: 1 }
> db.students.drop()
< true
studentDB >
```

3. Sample Student Data

```
> db.students.insertMany([
  { rollNo: 101, name: "John", age: 20, department: "CSE", marks: 85, city: "Hyderabad" },
  { rollNo: 102, name: "Alice", age: 21, department: "CSE", marks: 90, city: "Vijayawada" },
  { rollNo: 103, name: "Rahul", age: 20, department: "ECE", marks: 75, city: "Hyderabad" },
  { rollNo: 104, name: "Priya", age: 22, department: "IT", marks: 88, city: "Warangal" },
  { rollNo: 105, name: "Kiran", age: 21, department: "CSE", marks: 95, city: "Hyderabad" },
  { rollNo: 106, name: "Anu", age: 20, department: "IT", marks: 82, city: "Guntur" },
  { rollNo: 107, name: "Ravi", age: 23, department: "ECE", marks: 68, city: "Nellore" },
  { rollNo: 108, name: "Sita", age: 21, department: "CSE", marks: 91, city: "Hyderabad" },
  { rollNo: 109, name: "Arun", age: 22, department: "AI&DS", marks: 87, city: "Kurnool" },
  { rollNo: 110, name: "Meena", age: 20, department: "AI&DS", marks: 93, city: "Hyderabad" }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8e58b66aa2b06927cb5ebe'),
    '1': ObjectId('6a8e58b66aa2b06927cb5ebf'),
    '2': ObjectId('6a8e58b66aa2b06927cb5ec0'),
    '3': ObjectId('6a8e58b66aa2b06927cb5ec1'),
    '4': ObjectId('6a8e58b66aa2b06927cb5ec2'),
    '5': ObjectId('6a8e58b66aa2b06927cb5ec3'),
    '6': ObjectId('6a8e58b66aa2b06927cb5ec4'),
    '7': ObjectId('6a8e58b66aa2b06927cb5ec5'),
    '8': ObjectId('6a8e58b66aa2b06927cb5ec6'),
    '9': ObjectId('6a8e58b66aa2b06927cb5ec7')
  }
}
studentDB >
```

4. CREATE – Insert Commands

```
    }
  }
  < {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6a8e58b66aa2b06927cb5ebe'),
      '1': ObjectId('6a8e58b66aa2b06927cb5ebf'),
      '2': ObjectId('6a8e58b66aa2b06927cb5ec0'),
      '3': ObjectId('6a8e58b66aa2b06927cb5ec1'),
      '4': ObjectId('6a8e58b66aa2b06927cb5ec2'),
      '5': ObjectId('6a8e58b66aa2b06927cb5ec3'),
      '6': ObjectId('6a8e58b66aa2b06927cb5ec4'),
      '7': ObjectId('6a8e58b66aa2b06927cb5ec5'),
      '8': ObjectId('6a8e58b66aa2b06927cb5ec6'),
      '9': ObjectId('6a8e58b66aa2b06927cb5ec7')
    }
  }
}
> db.students.insertOne({
  rollNo: 111,
  name: "Vijay",
  age: 21,
  department: "CSE",
  marks: 86,
  city: "Hyderabad"
})

db.students.insertMany([
  { rollNo: 112, name: "Pooja", age: 20, department: "IT", marks: 84 },
  { rollNo: 113, name: "Ramesh", age: 22, department: "ECE", marks: 79 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8e590c6aa2b06927cb5ec9'),
    '1': ObjectId('6a8e590c6aa2b06927cb5eca')
  }
}
studentDB> |
```

5. READ – Basic Find Commands

```
< {  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6a8e590c6aa2b06927cb5ec9'),  
    '1': ObjectId('6a8e590c6aa2b06927cb5eca')  
  }  
}  
  
> db.students.find()  
db.students.findOne()  
db.students.findOne({ rollNo: 101 })  
db.students.find({ department: "CSE" })  
db.students.find({ name: "John" })  
db.students.countDocuments()  
db.students.countDocuments({ department: "CSE" })  
  
< 5  
studentDB > |
```

6. Comparison Operators

- Greater than

```
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec3'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec5'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec6'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec7'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
```

```
> db.students.find({ marks: { $gt: 80 } })
< {
  _id: ObjectId('6a8e58b66aa2b06927cb5e8e'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ebf'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec1'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec2'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
```

- Greater than or equal

```
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec3'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec5'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec6'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec7'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
```

```
> db.students.find({ marks: { $gte: 80 } })
< {
  _id: ObjectId('6a8e58b66aa2b06927cb5ebe'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ebf'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec1'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec2'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
```

```
{
  _id: ObjectId('6a8e590c6aa2b06927cb5ec8'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e590c6aa2b06927cb5ec9'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
studentDB> |
```

- Less than

```
> db.students.find({ marks: { $lt: 80 } })
< {
  _id: ObjectId('6a8e58b66aa2b06927cb5ec0'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec4'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8e590c6aa2b06927cb5eca'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
studentDB >
```

- Less than or equal

```
> db.students.find({ marks: { $lte: 80 } })
< {
  _id: ObjectId('6a8e58b66aa2b06927cb5ec0'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec4'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8e590c6aa2b06927cb5eca'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
studentDB >
```

- Not equal

```
> db.students.find({ department: { $ne: "CSE" } })
< {
  _id: ObjectId('6a8e58b66aa2b06927cb5ec0'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec1'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec3'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec4'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
```

```
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec6'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec7'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e590c6aa2b06927cb5ec9'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
{
  _id: ObjectId('6a8e590c6aa2b06927cb5eca'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
studentDB> |
```

- Equal

```
> db.students.find({ marks: { $eq: 90 } })
< {
  _id: ObjectId('6a8e58b66aa2b06927cb5ebf'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
studentDB>
```

7. Logical Operators

- AND

```
> db.students.find({
  $and: [
    { department: "CSE" },
    { marks: { $gt: 85 } }
  ]
})
< {
  _id: ObjectId('6a8e58b66aa2b06927cb5ebf'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec2'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec5'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e590c6aa2b06927cb5ec8'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
studentDB>
```

- OR

```
> db.students.find({
  $nor: [
    { department: "CSE" },
    { marks: { $gt: 90 } }
  ]
})
< {
  _id: ObjectId('6a8e58b66aa2b06927cb5ec0'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec1'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec3'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec4'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
```

```
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec4'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec6'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a8e590c6aa2b06927cb5ec9'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
{
  _id: ObjectId('6a8e590c6aa2b06927cb5eca'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
studentDB >
```

- NOT

```
> db.students.find({
  department: { $in: ["CSE", "IT"] }
})

db.students.find({
  department: { $nin: ["CSE", "IT"] }
})
< {
  _id: ObjectId('6a8e58b66aa2b06927cb5ec0'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec4'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8e58b66aa2b06927cb5ec6'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
```

```
_id: ObjectId('6a8e58b66aa2b06927cb5ec7'),
rollNo: 110,
name: 'Meena',
age: 20,
department: 'AI&DS',
marks: 93,
city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e590c6aa2b06927cb5eca'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
```

- NOR

```
> db.students.updateOne(
  { rollNo: 101 },
  { $set: { marks: 92 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

8. IN and NIN Operators

```
> db.students.updateMany(
  { department: "CSE" },
  { $set: { status: "Active" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

9. UPDATE Commands

- Update one

```
> db.students.updateOne(
  { rollNo: 101 },
  { $inc: { marks: 5 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> |
```

- Update multiple

```
> db.students.updateOne(
  { rollNo: 101 },
  { $inc: { marks: -2 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> |
```

- Increment

```
> db.students.updateMany(  
  {},  
  { $rename: { "city": "location" } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 13,  
  modifiedCount: 11,  
  upsertedCount: 0  
}
```

- Decrement

```
> db.students.updateOne(  
  { rollNo: 101 },  
  { $set: { attendance: 90 } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
studentDB> |
```

- Rename a field

```
> db.students.updateOne(  
  { rollNo: 101 },  
  { $unset: { attendance: "" } }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
studentDB> |
```

10. DELETE Commands

- Add field

```
> db.students.deleteOne({ rollNo: 103 })
< {
  acknowledged: true,
  deletedCount: 1
}
studentDB> |
```

- Remove a field

```
> db.students.deleteMany({ marks: { $lt: 40 } })
< {
  acknowledged: true,
  deletedCount: 0
}
studentDB> |
```

- Delete all documents

```
> db.students.deleteMany({})
< {
  acknowledged: true,
  deletedCount: 12
}
studentDB>
```

11. Sorting

```
> db.students.find().sort({ marks: 1 })
<
> db.students.find().sort({ marks: -1 })
<
> db.students.find().sort({ department: 1, marks: -1 })
<
studentDB>
```

12. Limit and Skip

```
> db.students.find().limit(5)
<
> db.students.find().sort({ marks: -1 }).limit(3)
<
> db.students.find().skip(5)
<
> db.students.find().skip(2).limit(3)
<
studentDB> |
```

13. Projection

- Only name and marks

```
> db.students.find(
  {},
  { _id: 0, name: 1, marks: 1 }
)
<
studentDB> |
```

- Name, department and marks

```
> db.students.find(
  {},
  { _id: 0, name: 1, department: 1, marks: 1 }
)
<
studentDB> |
```

- Hide age

```
> db.students.find(
  {},
  { age: 0 }
)
<
studentDB> |
```

14. Distinct Values

```
> db.students.distinct("department")
< []
> db.students.distinct("city")
< []
> db.students.distinct(
  "department",
  { marks: { $gte: 80 } }
)
< []
studentDB>
```

15. Index Commands

- Create index

```
db.students.createIndex({ rollNo: 1 })
rollNo_1
```

- Create name index

```
db.students.createIndex({ name: 1 })
name_1
```

- Compound index

```
db.students.createIndex({ department: 1, marks: -1 })
department_1_marks_-1
```

- Show indexes

```
db.students.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { rollNo: 1 }, name: 'rollNo_1' },
  { v: 2, key: { name: 1 }, name: 'name_1' },
  {
    v: 2,
    key: { department: 1, marks: -1 },
    name: 'department_1_marks_-1'
  }
]
```

- Drop name index

```
db.students.dropIndex("name_1")
{ nIndexesWas: 4, ok: 1 }
```

- Drop all indexes except _id

```
> db.students.dropIndexes()
< {
  nIndexesWas: 3,
  msg: 'non-_id indexes dropped for collection',
  ok: 1
}
studentDB>
```

16. Aggregation – Basic Commands

- Display all documents through aggregation

```
> db.students.distinct("department")
< []
> db.students.aggregate([
  { $match: {} }
])
<
studentDB> |
```

- Students with marks > 80

```

    })
  <
  > db.students.aggregate([
    { $match: { marks: { $gt: 80 } } }
  ])
  <
  studentDB> |

```

- Department-wise student count

```

  <
  > db.students.aggregate([
    {
      $group: {
        _id: "$department",
        totalStudents: { $sum: 1 }
      }
    }
  ])
  <
  studentDB> |

```

- Department-wise average marks

```

  > db.students.aggregate([
    {
      $group: {
        _id: "$department",
        averageMarks: { $avg: "$marks" }
      }
    }
  ])
  <
  studentDB>

```

- Highest marks

```

  > db.students.aggregate([
    {
      $group: {
        _id: null,
        highestMarks: { $max: "$marks" }
      }
    }
  ])
  <
  studentDB>

```

- Lowest marks

```
> db.students.aggregate([
  {
    $group: {
      _id: null,
      lowestMarks: { $min: "$marks" }
    }
  }
])
<
studentDB>
```

- Total marks

```
> db.students.aggregate([
  {
    $group: {
      _id: null,
      totalMarks: { $sum: "$marks" }
    }
  }
])
<
studentDB>
```

17. Aggregation with Multiple Stages

- CSE students, grouped and sorted by average marks

```
> db.students.aggregate([
  { $match: { department: "CSE" } },
  {
    $group: {
      _id: "$department",
      averageMarks: { $avg: "$marks" },
      totalStudents: { $sum: 1 }
    }
  },
  { $sort: { averageMarks: -1 } }
])
```

- Top 3 students using aggregation

```

> db.students.aggregate([
  { $sort: { marks: -1 } },
  { $limit: 3 },
  {
    $project: {
      _id: 0,
      rollNo: 1,
      name: 1,
      marks: 1
    }
  }
])

```

18. Useful MongoDB Commands

- show dbs
- show collections
- db
- db.students.find().pretty()
- db.students.countDocuments()
- db.students.distinct("department")
- db.students.getIndexes()

```

> show dbs
< admin      112.00 KiB
  config     88.00 KiB
  local      64.00 KiB
  studentDB  24.00 KiB
studentDB>

```

```

> show collections
< students
studentDB>

```

```

> db.students.find().pretty()
<
studentDB>

```

```

> db.students.countDocuments()
< 0

```

```
> db.students.distinct("department")
< []
studentDB>
```

```
> db.students.getIndexes()
< [ { v: 2, key: { _id: 1 }, name: '_id_' } ]
studentDB>
```

19. Complete CRUD Practical

- CREATE

```
> db.students.insertOne({
  rollNo: 201,
  name: "Test Student",
  age: 20,
  department: "CSE",
  marks: 85
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a91159528b4dbe03e09a5e2')
}
```

- READ

```
> db.students.updateOne(
  { rollNo: 201 },
  { $set: { marks: 95 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

DELETE

```
> db.students.deleteOne({ rollNo: 201 })
< {
  acknowledged: true,
  deletedCount: 1
}
studentDB>
```